

INSERTION SORT PROGRAM

```
#include<stdio.h>

# define size 20

void insertion_sort(int arr[] , int n )
{
    for(int i = 1 ; i <= n-1 ; i++)
    {
        int temp = arr[i];
        int j = i-1;
        for( ; j>=0 ; j-- )
        {
            if(arr[j] > temp
                arr[j+1]=arr[j];
            else
                break;
        }
        arr[j+1] = temp;
    }
    for(int i = 0 ; i < n ; i++)
    {
        printf("%d,",arr[i]);
    }
}

int main()
{
    int arr[size] , n ;
    printf("\n enter the number of element in the array = ");
    scanf("%d",&n);
    printf("\n enter the data in the array : ");
    for(int i = 0 ; i < n ; i++)
    {
```

```

        printf("\n enter the arr[%d] element : ",i+1);
        scanf("%d",&arr[i]);
    }
    insertion_sort(arr , n);
    printf("\n now the sorted array is ... [");
    for(int i = 0 ; i < n ; i++)
    {
        printf("%d,",arr[i]);
    }
    printf("]");
}

```

OUTPUT :

enter the number of element in the array = 5

enter the data in the array :

enter the arr[1] element : 4

enter the arr[2] element : 3

enter the arr[3] element : 5

enter the arr[4] element : 2

enter the arr[5] element : 1

now the sorted array is ... [1,2,3,4,5,]

ALGORITHM:

INSERTIONSORT (ARR, N)

Step 1: Repeat Steps 2 to 5 for K = 1 to N – 1

Step 2: SET TEMP = ARR[K]

Step 3: SET J = K - 1

Step 4: Repeat while TEMP <= ARR[J]

 SET ARR[J + 1] = ARR[J]

 SET J = J - 1

 [END OF INNER LOOP]

Step 5: SET ARR[J + 1] = TEMP

 [END OF LOOP]

Step 6: EXIT